

CSE4203: Computer Graphics
Lecture - 4 (Part - A)

Geometric Transformation

Outline

- Transformation
- Linear Transformation
 - Scaling
 - Shearing
 - Rotation
- Composite Transformation

2D Linear Transformations (1/1)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y \\ a_{21}x + a_{22}y \end{bmatrix}$$

Linear Transformation: Operation of taking a **vector** and produces **another vector** by a **simple matrix multiplication**.

Scaling (1/6)

- The most basic transform is a scale along the coordinate axes.

$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

- s_x and s_y is called the **scaling factors** which determines how much scaling is applied along the x and y axis

Scaling (2/6)

- The most basic transform is a scale along the coordinate axes.

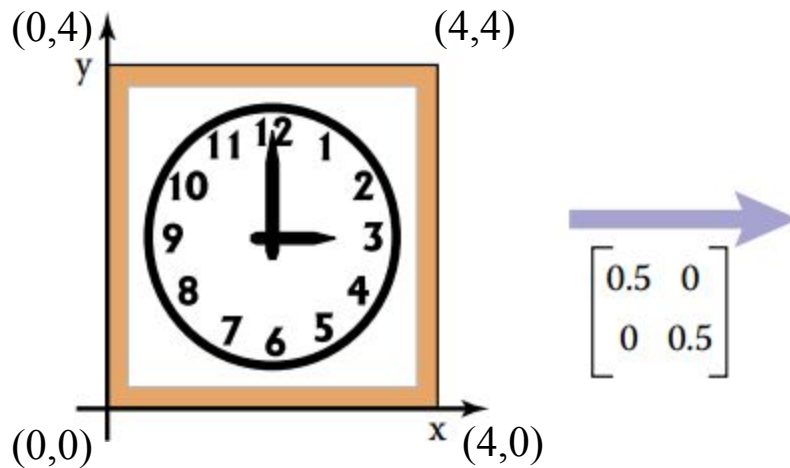
$$\text{scale}(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

The matrix does to a vector with Cartesian components (x, y):

$$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} s_x x \\ s_y y \end{bmatrix}$$

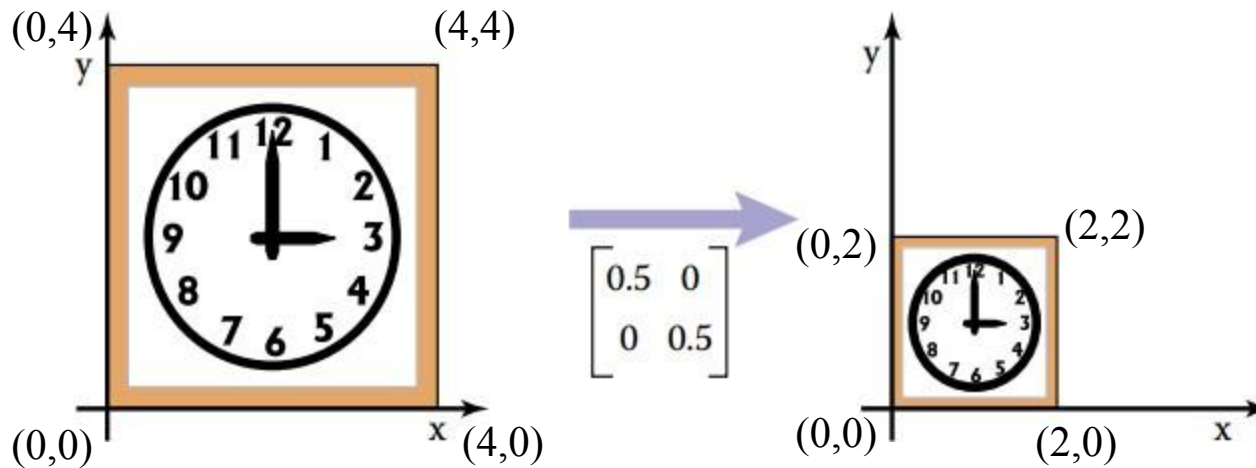
Scaling (3/6)

$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



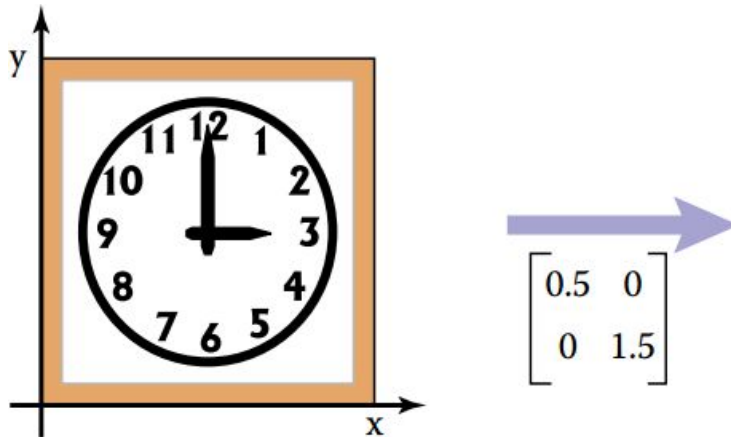
Scaling (4/6)

$$\text{scale}(0.5, 0.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix}$$



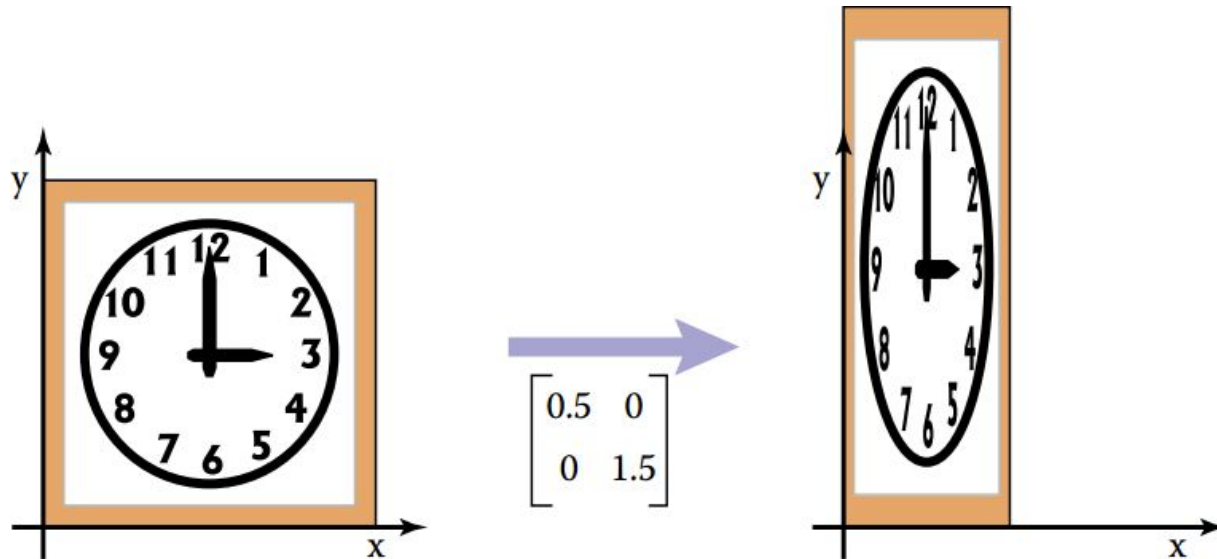
Scaling (5/6)

$$\text{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



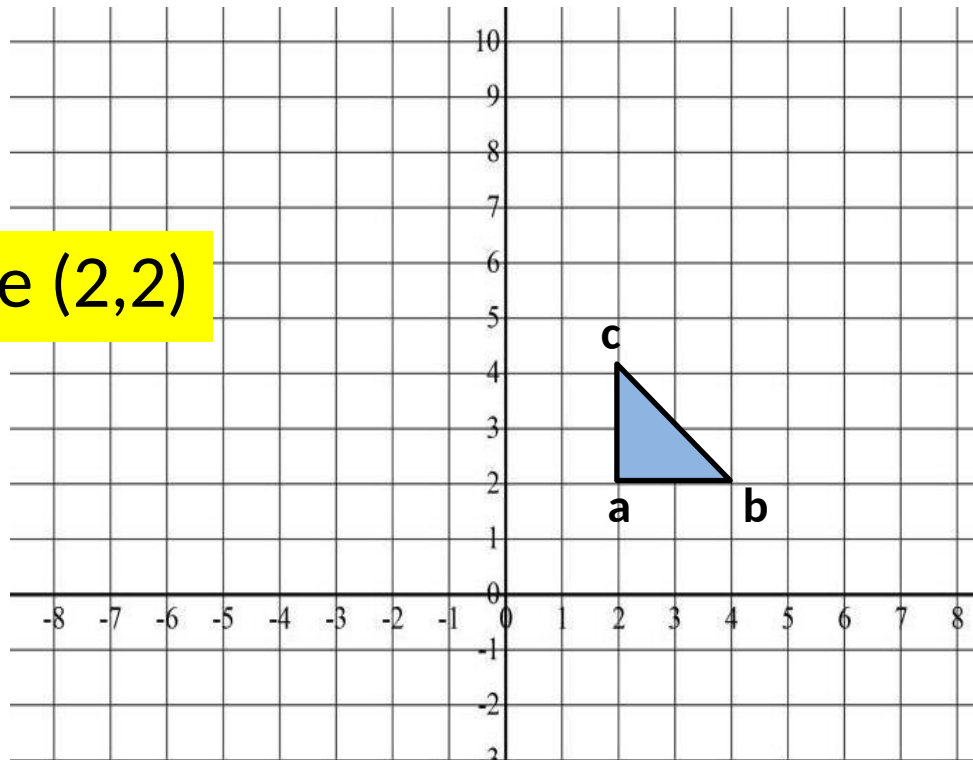
Scaling (6/6)

$$\text{scale}(0.5, 1.5) = \begin{bmatrix} 0.5 & 0 \\ 0 & 1.5 \end{bmatrix}$$



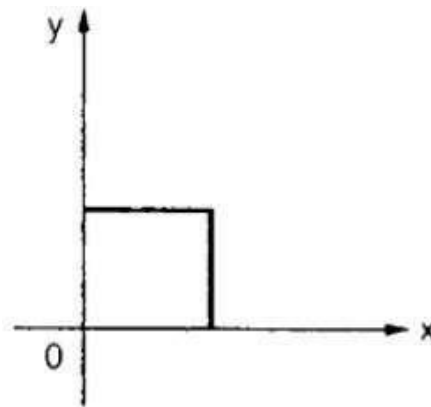
Scaling (6/6)

Scale (2,2)

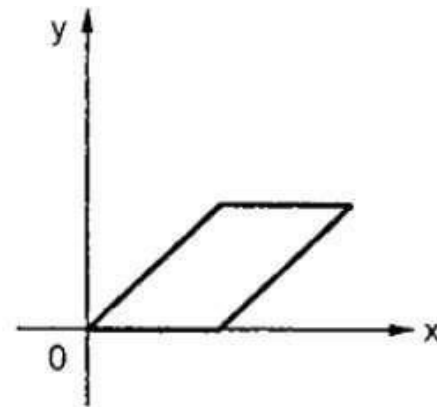


Shearing (1/5)

- A shear is something that pushes things sideways



(a) Original object



(b) Object after x shear

Shearing (2/5)

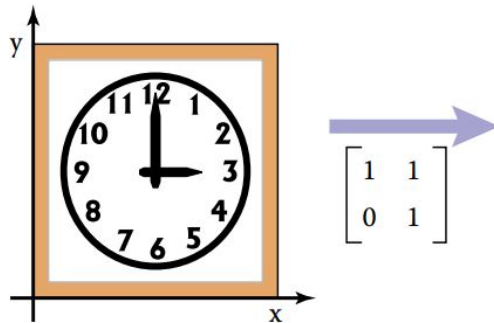
- A shear is something that pushes things sideways



$$\text{shear-x}(s) = \begin{bmatrix} 1 & s \\ 0 & 1 \end{bmatrix}, \quad \text{shear-y}(s) = \begin{bmatrix} 1 & 0 \\ s & 1 \end{bmatrix}$$

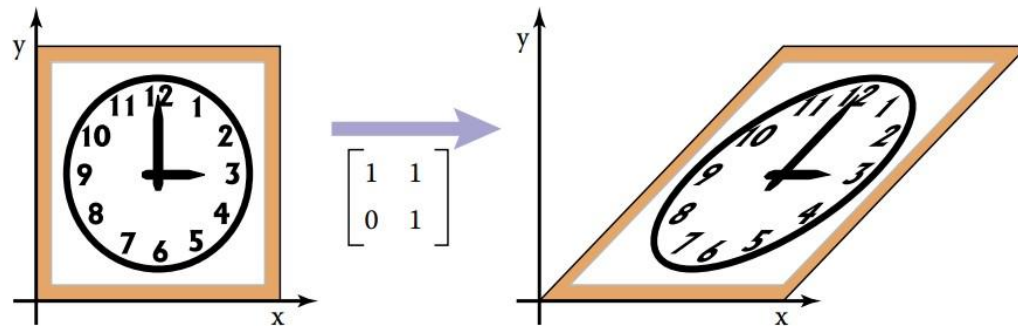
Shearing (3/5)

$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



Shearing (4/5)

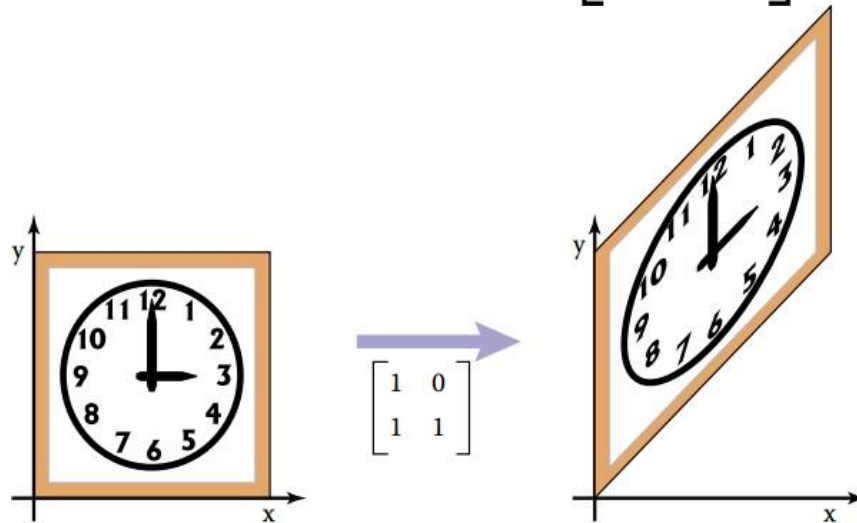
$$\text{shear-x}(1) = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$



An x-shear matrix moves points to the right in proportion to their y-coordinate.

Shearing (5/5)

$$\text{shear-y}(1) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

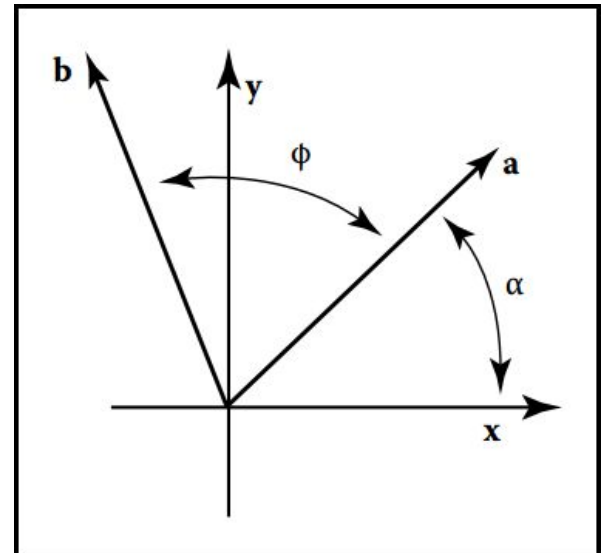


A y-shear matrix moves points up in proportion to their x-coordinate

Rotation (1/11)

$$x_a = r \cos \alpha,$$

$$y_a =$$



Rotation (2/11)

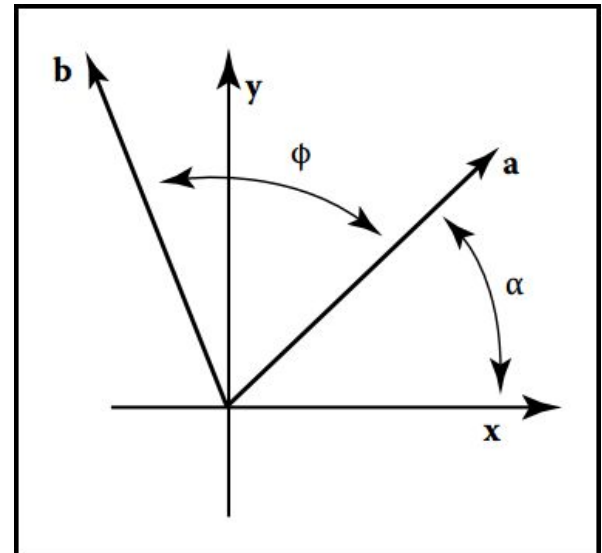
$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

$$x_b = r \cos(\alpha + \phi) =$$

$$y_b =$$

Anti-clockwise rotation = (+) rotation
Clockwise rotation = (-) rotation



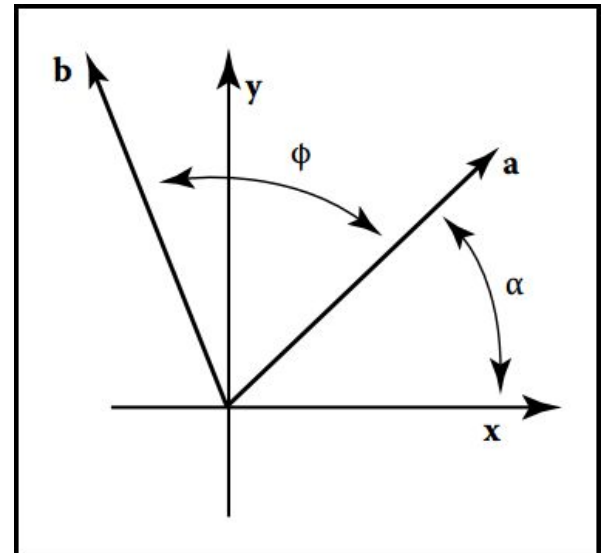
Rotation (3/11)

$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b =$$



Rotation (4/11)

$$x_a = r \cos \alpha,$$

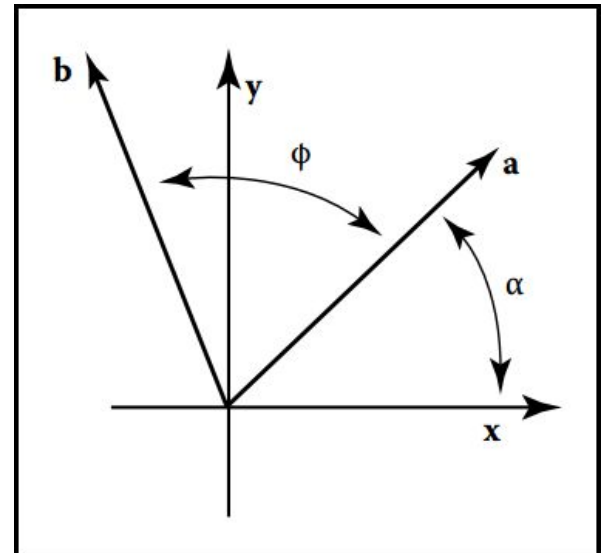
$$y_a = r \sin \alpha.$$

$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b = r \sin(\alpha + \phi) = r \sin \alpha \cos \phi + r \cos \alpha \sin \phi.$$

$$x_b =$$

$$y_b =$$



Rotation (5/11)

$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

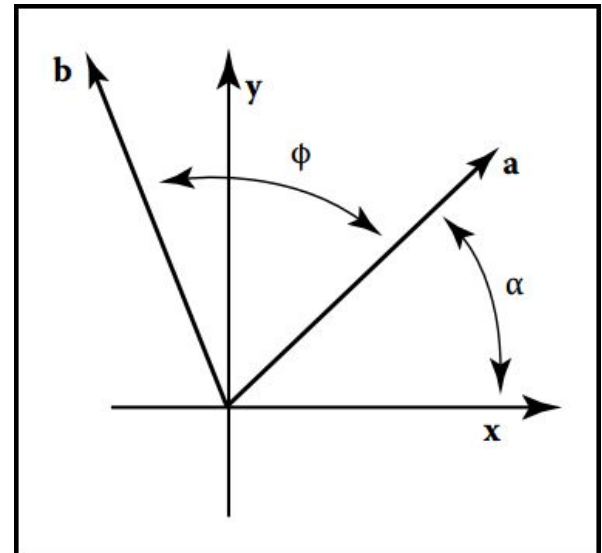
$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b = r \sin(\alpha + \phi) = r \sin \alpha \cos \phi + r \cos \alpha \sin \phi.$$

$$x_b = x_a \cos \phi - y_a \sin \phi,$$

$$y_b = y_a \cos \phi + x_a \sin \phi.$$

$$\text{rotate}(\phi) = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix}$$



Rotation (6/11)

$$x_a = r \cos \alpha,$$

$$y_a = r \sin \alpha.$$

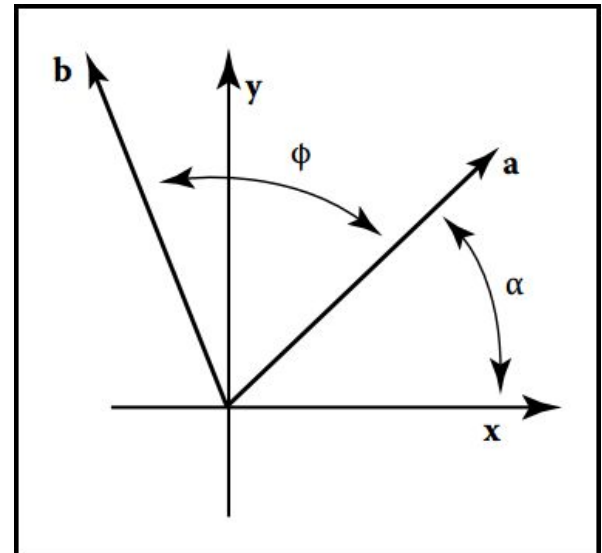
$$x_b = r \cos(\alpha + \phi) = r \cos \alpha \cos \phi - r \sin \alpha \sin \phi,$$

$$y_b = r \sin(\alpha + \phi) = r \sin \alpha \cos \phi + r \cos \alpha \sin \phi.$$

$$x_b = x_a \cos \phi - y_a \sin \phi,$$

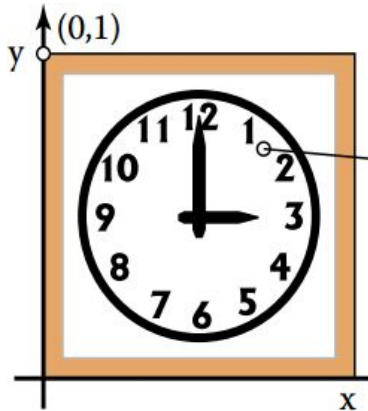
$$y_b = y_a \cos \phi + x_a \sin \phi.$$

$$\text{rotate}(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$



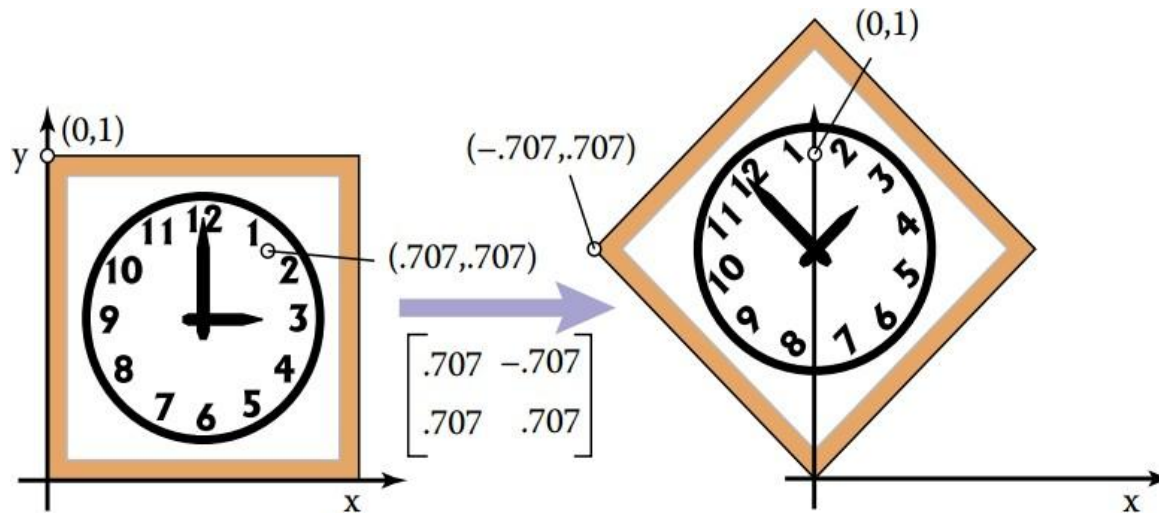
Rotation (7/11)

$$\begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$



Rotation (8/11)

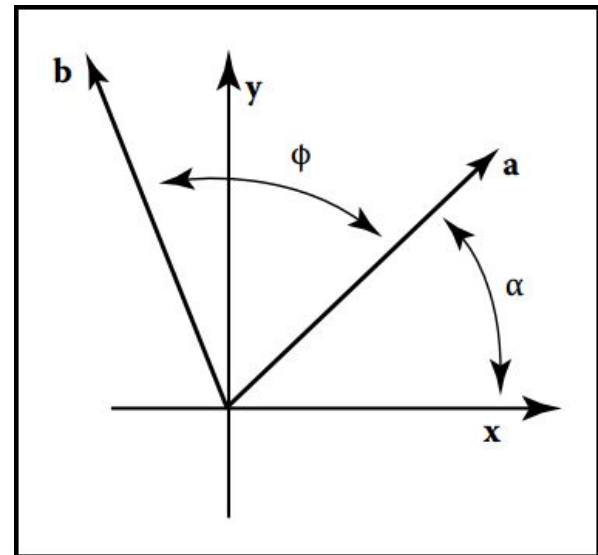
$$\begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$



Rotation (9/11)

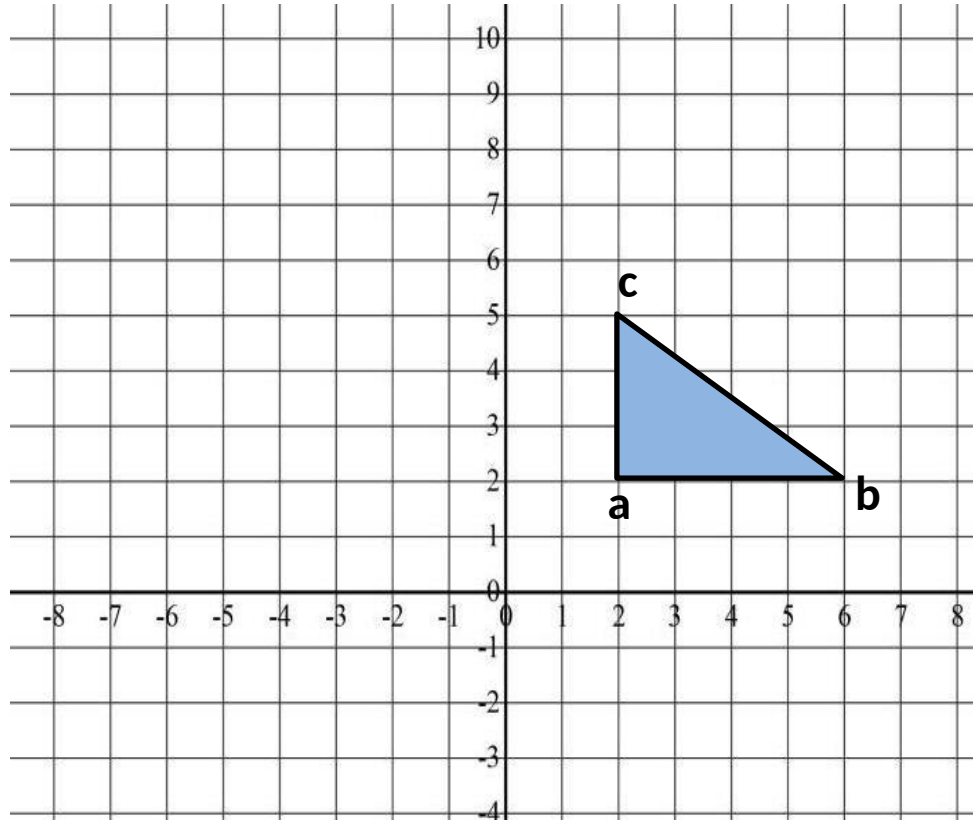
$$\text{rotate}(\phi) = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}$$

Q: What about - ve angle?



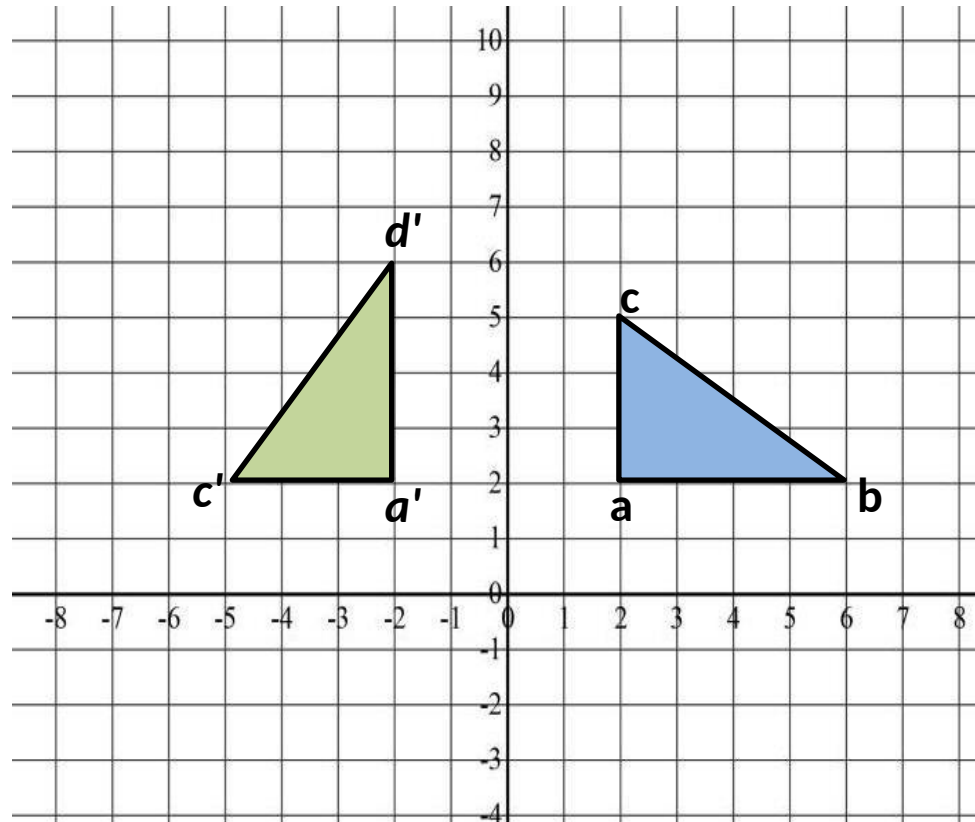
Rotation (10/11)

Rotate(90)



Rotation (11/11)

Rotate(90)

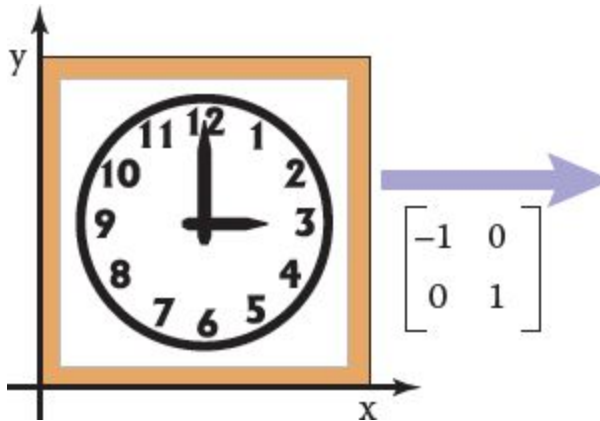


Reflection (1/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

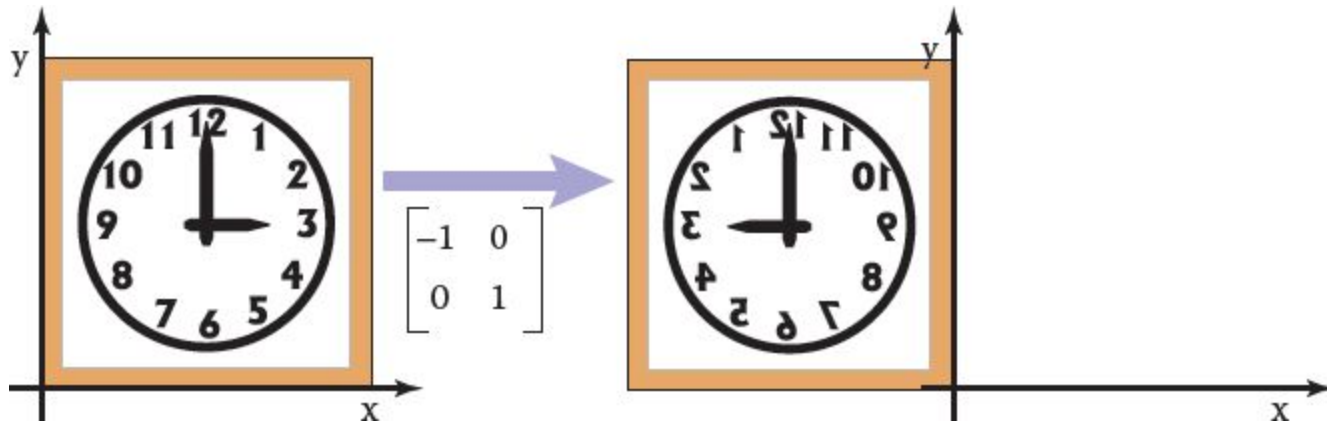
Reflection (2/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



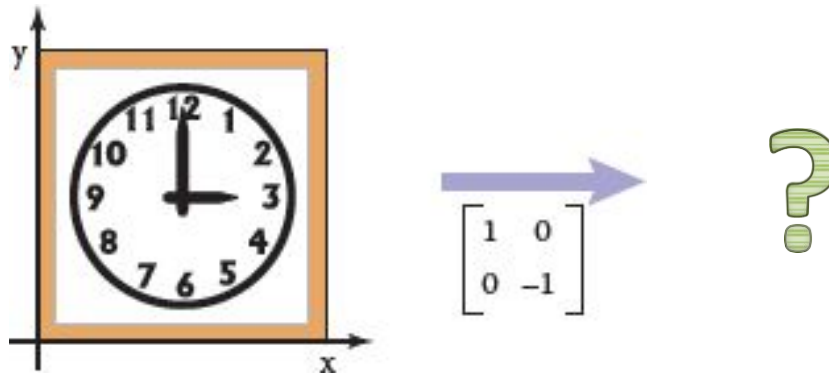
Reflection (3/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



Reflection (5/5)

$$\text{reflect-y} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \text{reflect-x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



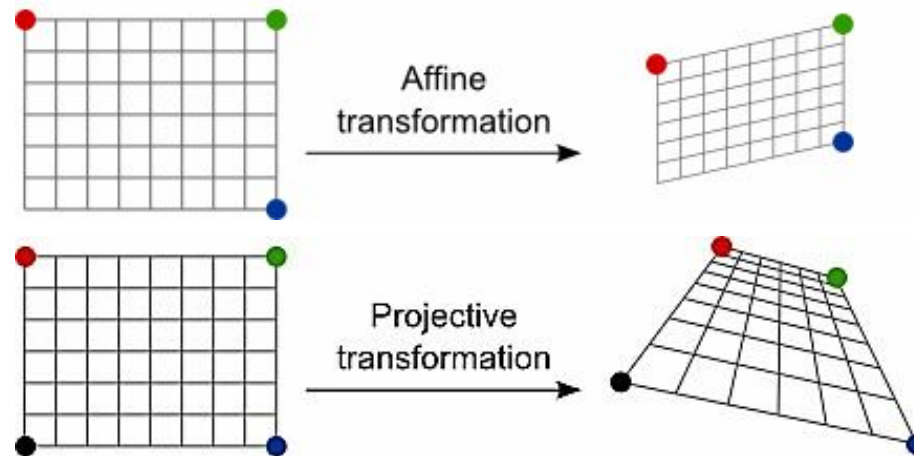
Affine transformation (1/2)

- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.

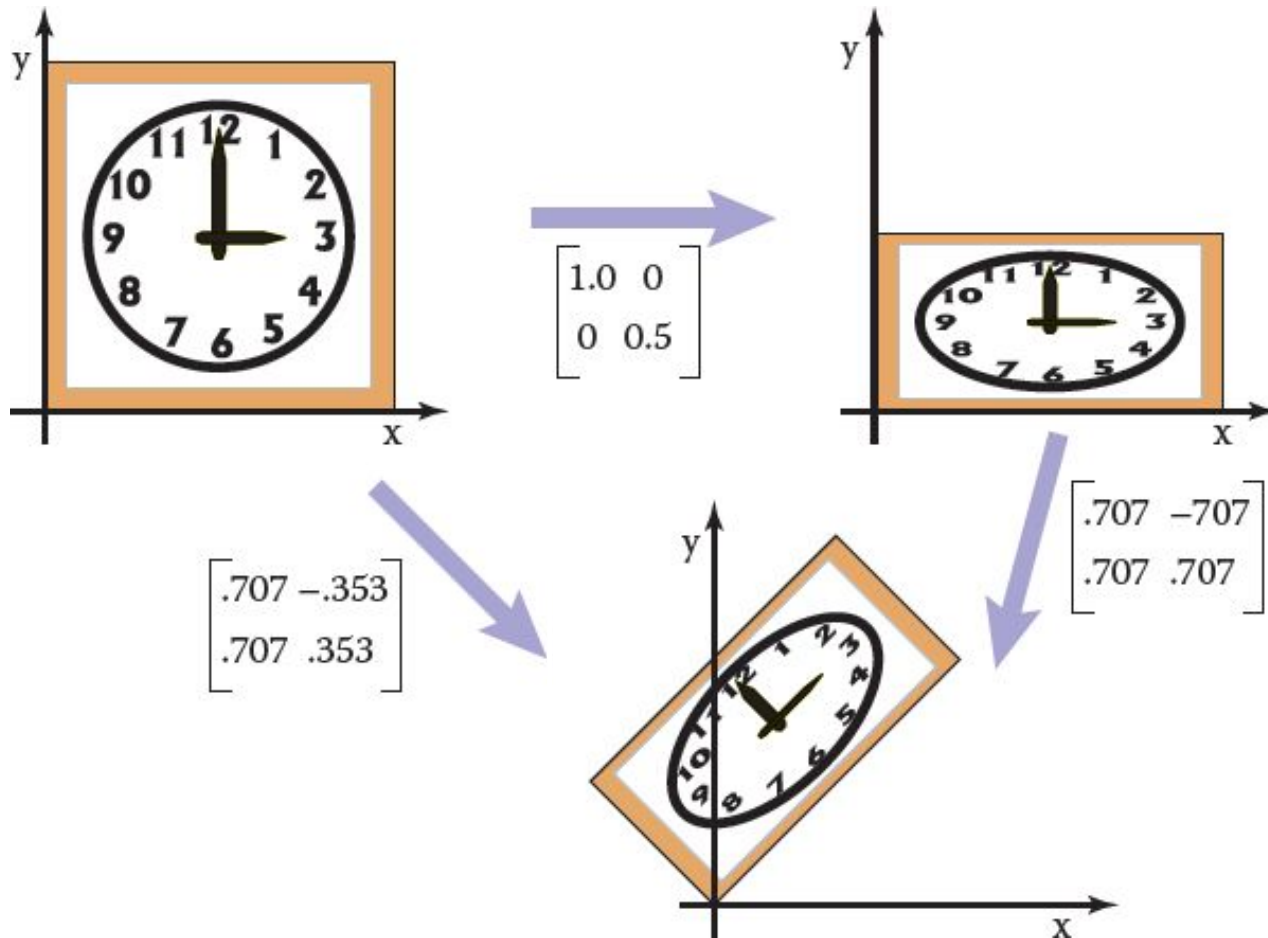


Affine transformation (2/2)

- Maps points to points, lines to lines, planes to planes.
- Preserves the ratio of lengths of parallel line segments.
- Sets of parallel lines remain parallel.
- Does not necessarily preserve angles between lines or distances between points.



Composition of Transformations (1/11)



Composition of Transformations (2/11)

- Composite transformation is the process of combining multiple transformations
 - i.e., for a 2D point v_1 we might want to –
 1. first apply a scale S
 2. then a rotation R .
- This would be done in two steps:
 1. **first, $v_2 = S v_1$**
 2. **then, $v_3 = R v_2$.**

Composition of Transformations (3/11)

Therefore –

1. $v_2 = S v_1$

2. $v_3 = R v_2$

3. $v_3 = R (Sv_1)$

4. $v_3 = (RS) v_1$ [*matrix multiplication is associative*]

5. $v_3 = Mv_1$ [*Where $M=RS$*]

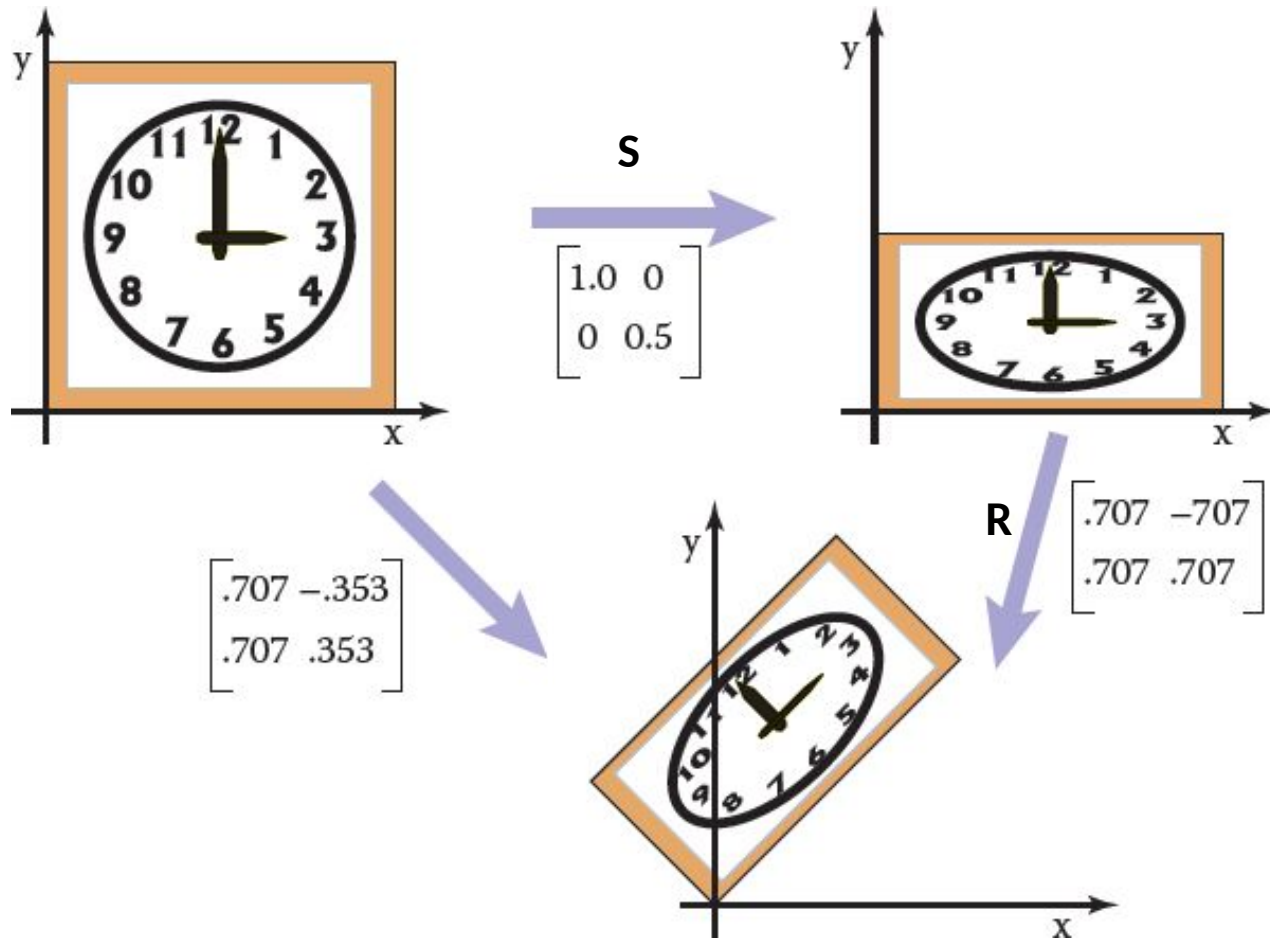
Composition of Transformations (4/11)

$$\mathbf{v}' = \mathbf{M} \mathbf{v}$$

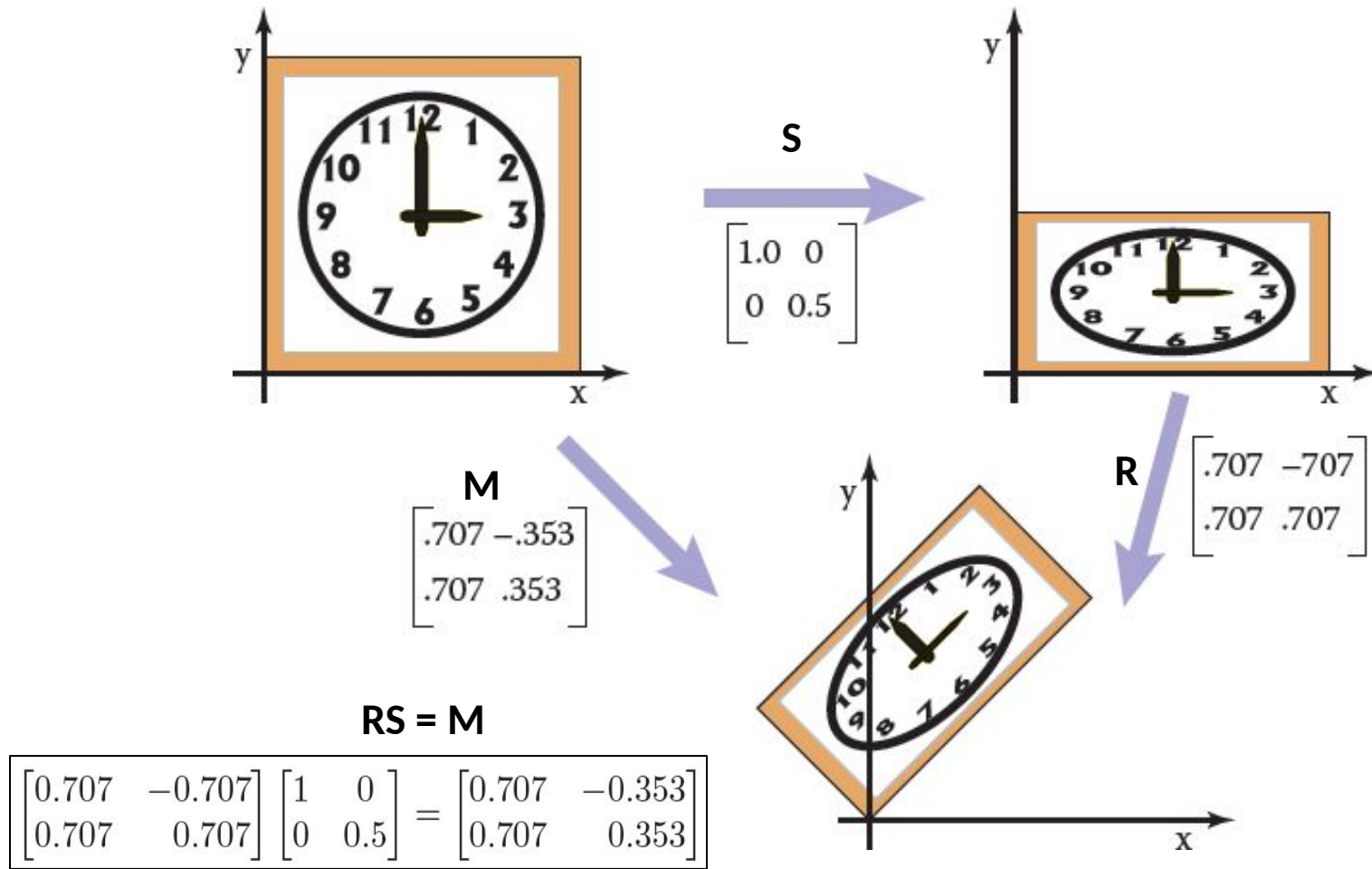
[Where $\mathbf{M} = \mathbf{R} \mathbf{S}$]

- We can represent the effects of transforming a vector by two matrices in sequence using a single matrix of the same size
 - which we can compute by multiplying the two matrices: $\mathbf{M} = \mathbf{R} \mathbf{S}$

Composition of Transformations (6/11)



Composition of Transformations (7/11)



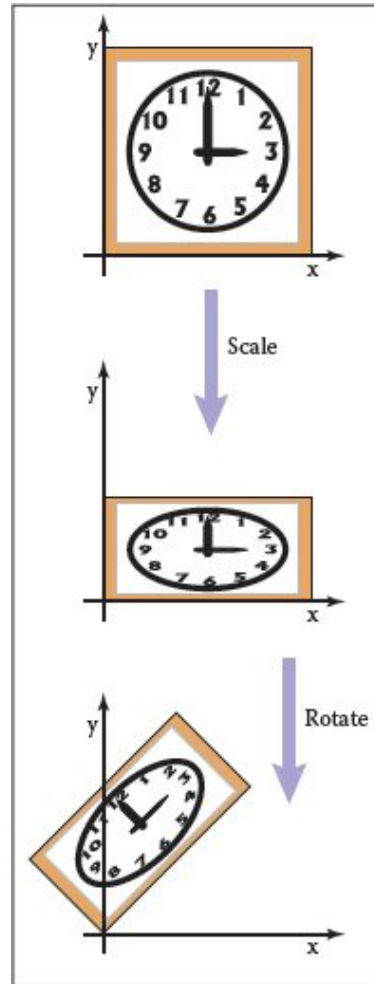
Credit: Fundamentals of Computer Graphics 3rd Edition by Peter Shirley, Steve Marschner | <http://www.cs.cornell.edu/courses/cs4620/2019fa/>

Composition of Transformations (8/11)

- It is very important to remember that these transforms are applied :
 - **from the right side first.**
 - So the matrix $M = RS$
 - first applies S and then R .

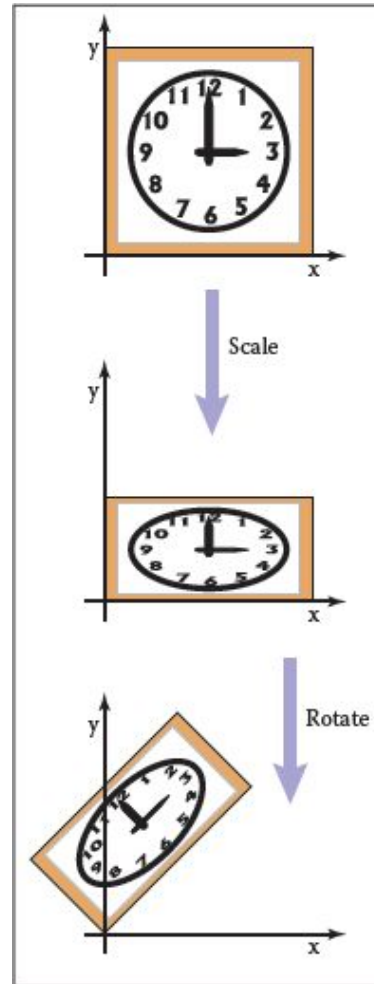
Composition of Transformations (9/11)

$$M = RS$$

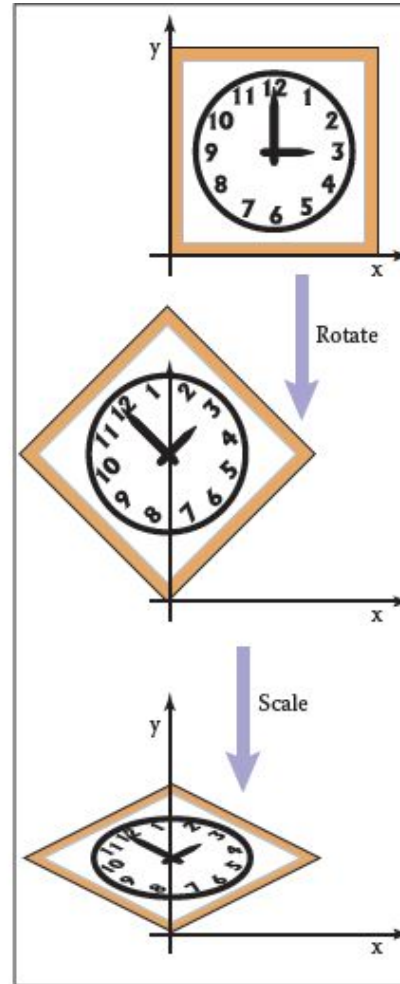


Composition of Transformations (10/11)

$$M = RS$$

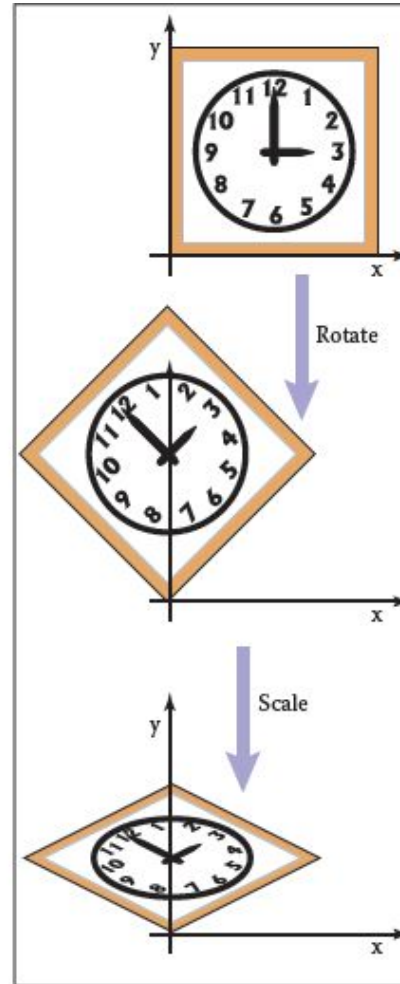
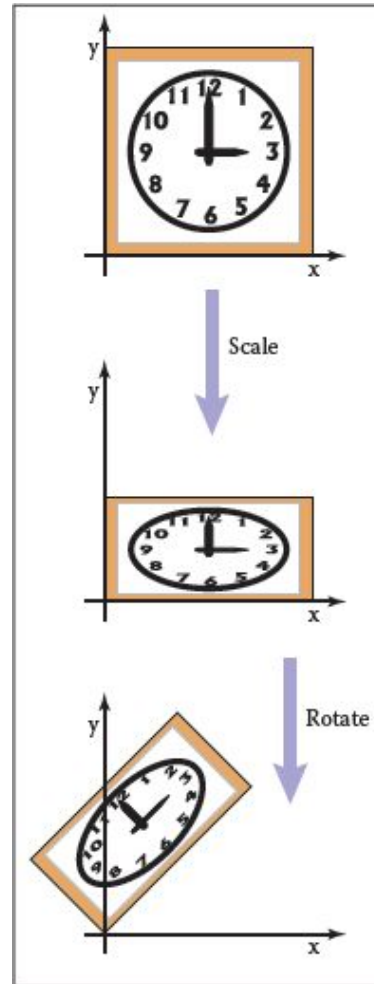


$$M = ?$$



Composition of Transformations (11/11)

$$M = RS$$

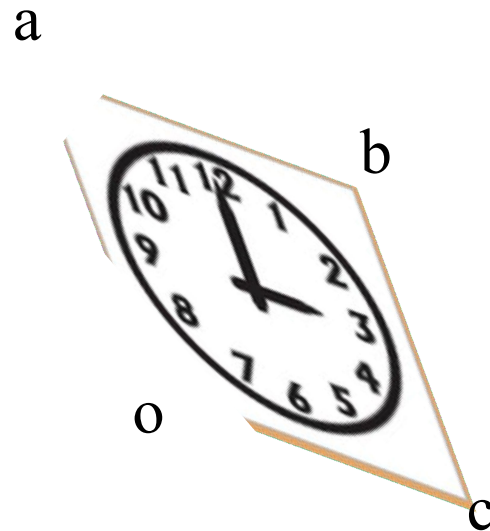
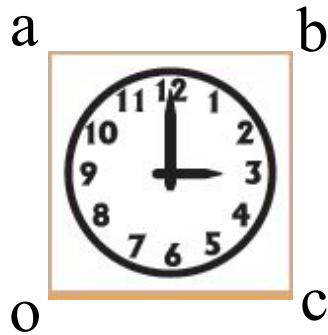


$$M =$$

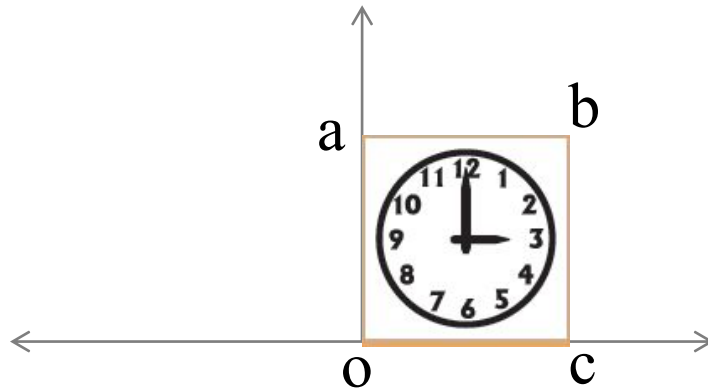
Q: What about more
than two
transformations:
 $T_1 \rightarrow T_2 \rightarrow T_3 \dots$
 $\rightarrow T_n$

Practice Problem - 1

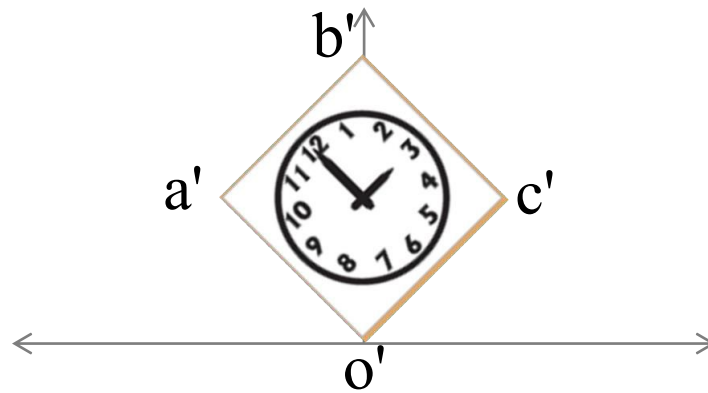
- Stretch the clock by 50% along one of its diagonals
 - so that 8:00 through 1:00 move to the northwest and 2:00 through 7:00 move to the southeast.



Practice Problem – 1 (Sol.)

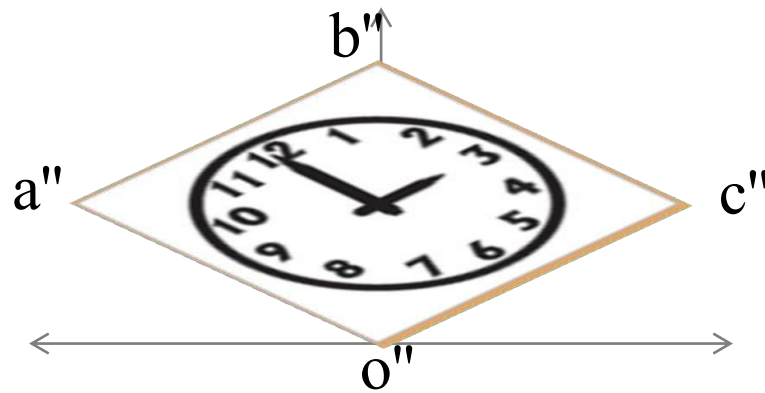


Practice Problem – 1 (Sol.)



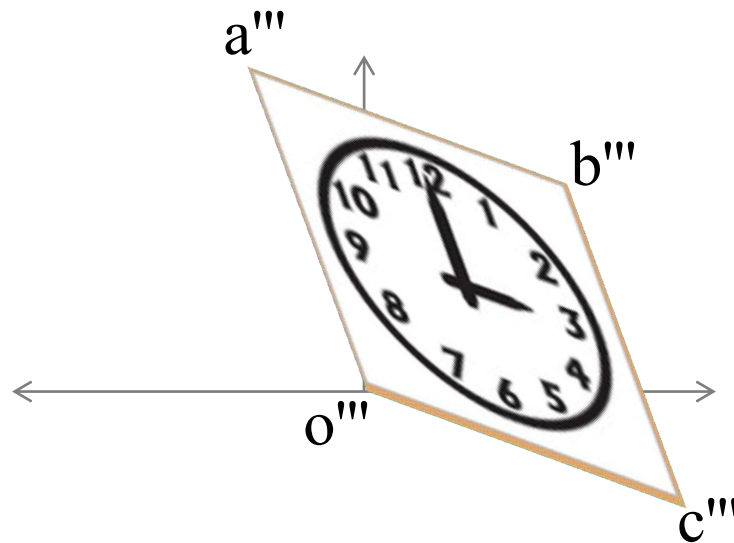
rotate(?)

Practice Problem – 1 (Sol.)



scale(?)

Practice Problem – 1 (Sol.)



rotate(?)

Practice Problem – 1 (Sol.)

- rotate(45°) $\rightarrow$ scale(1.5, 1) $\rightarrow$ rotate(-45°).

– *Q: Draw the steps*

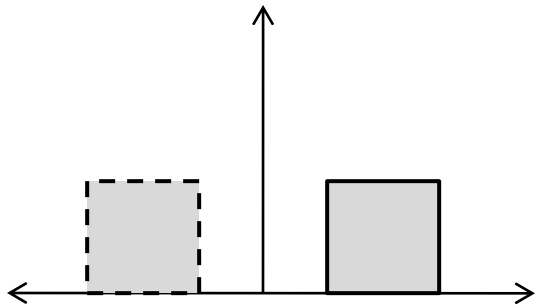
- $M = R(-45^\circ) S(1.5, 1) R(45^\circ)$
 $= R^T S R$

– *Q: Calculate the matrix*

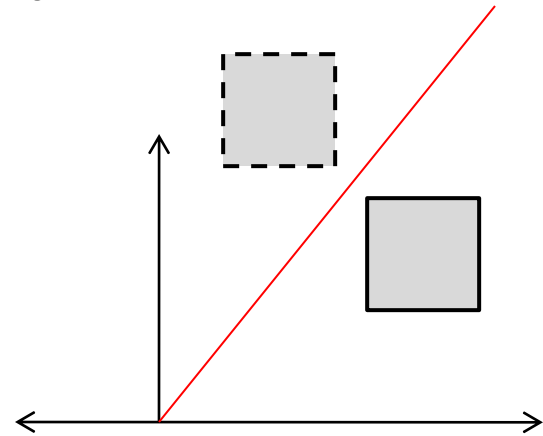
Practice Problem – 2

- Reflect the clock along a line goes through origin:

$$y = mx + c$$



w.r.t y-axis



w.r.t arbitrary line

Further Reading

- Fundamentals of Computer Graphics, 4th Edition -
Chapter 6

Thank you